Bloodstain Pattern Analysis

A Multidisciplinary Review of Fluid Dynamics, Machine Learning, and Image Processing Approaches

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May 2026

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Outline

What is Bloodstain Pattern Analysis (BPA)?

Definition

BPA is a forensic subspecialty — borrowing from **chemistry, biology, and physics** — concerned with the evaluation and interpretation of bloodstain patterns found at crime scenes.

Why it matters

Accurate BPA can confirm or **refute** a suspect's statement about the position of the victim during an assault — critical forensic testimony.

Core objectives:

- Determine the impact angle of each spatter stain
- Reconstruct the region of origin in 3-D space
- Classify pattern types objectively
- Link pattern evidence to crime reconstruction

Four Papers Reviewed

[1] Joris et al. (2014) – ABSM
[2] Choudhury et al. (2023) – ANN / $\beta_{\max}$
[3] Attinger et al. (2019) – Fluid dynamics & stats
[4] Arthur et al. (2017) – Image processing

The Physical Picture: Drop Impact

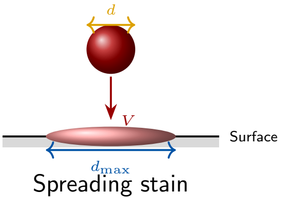
Stages of droplet impingement (Choudhury et al., 2023):

1. **Kinematic** — initial contact
2. **Spreading** — inertia drives lateral expansion
3. **Relaxation** — surface tension and viscosity resist
4. **Wetting/Equilibrium** — final resting state

The **maximum spreading ratio** $\beta_{\max}$ is the key dimensionless descriptor:

$$\beta_{\max} = \frac{d_{\max}}{d}$$

where $d_{\max}$ is the maximum spread diameter and d is the initial droplet diameter.



Key Dimensionless Numbers

Weber Number (We)

Ratio of inertial to surface-tension forces:

$$We = \frac{\rho V^2 d}{\sigma}$$

High $We \Rightarrow$ spreading dominates.

Reynolds Number (Re)

Ratio of inertial to viscous forces:

$$Re = \frac{\rho V d}{\mu}$$

High $Re \Rightarrow$ less viscous resistance.

Ohnesorge Number (Oh)

$$Oh = \frac{\mu}{\sqrt{\rho d \sigma}} = \frac{\sqrt{We}}{Re}$$

Scales viscous and surface-tension forces relative to inertia.

Impact Angle α

Classical ellipse-based formula (Balthazard, 1939):

$$\alpha \approx \arcsin\left(\frac{W}{L}\right)$$

W = stain width, L = stain length.

Motivation: Prior Models for $\beta_{\max}$

Model	**Formula**	**Condition**
Jones (1971)	$\beta_{\max} = \left(\frac{4}{3} Re^{1/4}\right)^{1/2}$	Viscous force only
Collings et al. (1990)	$\beta_{\max} = \left(\frac{We}{6}\right)^{1/2}$	Viscosity ignored
Madejski (1976)	$\frac{3\beta_{\max}^2}{We} + \frac{1}{Re}\left(\frac{\beta_{\max}}{1.2941}\right)^5 = 1$	$Re > 100,\ We > 100$
Pasandideh-Fard et al.	$\beta_{\max} = \sqrt{\frac{We + 12}{3(1 - \cos\theta) + 4We/\sqrt{Re}}}$	Fixed spreading time
Choudhury et al. (ANN)	**Eq. (9)**	**General**

All prior models fail in the range $\beta_{\max} \in [3, 8]$ and ignore surface roughness r_a.

Energy Balance Framework

The general energy conservation at impact (Choudhury et al., 2023):

$$\underbrace{E_{K,i} + E_{L,i} + E_{S,i}}_{\text{before impact}} = \underbrace{E_{K,f} + E_{L,f} + E_{S,f}}_{\text{at max spread}} + E_{\text{visc}}$$

- E_K — kinetic energy of drop
- E_L — free surface energy (surface tension γ_{LV})
- E_S — interfacial energy (liquid–surface)
- E_{visc} — viscous dissipation

Key limitation of analytic models

Setting up boundary conditions for E_{visc} requires assumptions about flow geometry (slug / coin-shaped disk / boundary layer thickness) that *break down* at low Re or on rough surfaces.

ANN Architecture (Choudhury et al., 2023)

Structure: 4-4-1 network (4 inputs, 4 hidden neurons, 1 output).

Inputs: Re, We, θ_{adv}, r_a
Output: $\beta_{\max}$

Hidden-layer activation (sigmoid):

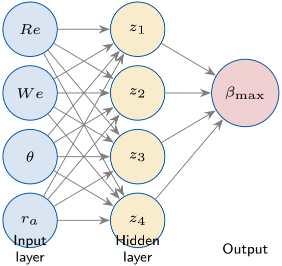
$$z_j(x) = \frac{2}{1 + e^{-2a_j}} - 1$$

where $a_j(x) = \sum_{i=1}^{4} w_{ji} x_i + C_j$

Output:

$$o_k(x) = \sum_{j=1}^{4} w_{jk}\, z_j + d_k$$

Training: Levenberg–Marquardt



Normalisation and Final ANN Equation

Input normalisation (prevents scale-induced bias):

$$\bar{x} = \frac{2(x - x_{\min})}{x_{\max} - x_{\min}} - 1 \in [-1, 1]$$

Trained ANN equation for $\beta_{\max}$ (Choudhury et al., 2023, Eq. 9):

$$\begin{aligned}\beta_{\max} = \Bigg[&0.3601\left(\frac{2}{1 + e^{-2(-4.994Re+3.162We+8.644r_a+3.432\theta+8.834)}} - 1\right)\\ &- 2.0401\left(\frac{2}{1 + e^{-2(-1.432Re+0.205We-2.603r_a-1.433\theta+3.759)}} - 1\right)\\ &- 0.4037\left(\frac{2}{1 + e^{-2(-2.762Re+2.216We+8.644r_a+5.613\theta+6.711)}} - 1\right)\\ &- 4.3533\left(\frac{2}{1 + e^{-2(-0.921Re-0.436We-0.049r_a-0.003\theta-2.442)}} - 1\right)\Bigg] - 2.4024\end{aligned}$$

ANN Training Parameters and Dataset

Table: ANN hyperparameters

Parameter	Value
Max epochs	5000
Max val. failures	20
Learning rate	0.01
Min gradient	10^{-10}
Initial μ (LM)	1
μ decrease factor	0.8
μ increase factor	1.5

Table: Dataset summary (342 points)

Variable	Range
We	1.1–2055
Re	9–$15{,}860$
θ (rad)	0.10–2.83
r_a (nm)	1.3–6200
d_{droplet} (mm)	0.003–5

Sources

Data aggregated from 12 independent studies including Choudhury et al. (2017), Lee et al. (2016), Rioboo et al. (2002), Antonini et al. (2012), and others.

ANN Results: Model Comparison

Table: Blind-test performance (R^2 / MSE)

Model	R^2	MSE
Madejski (1976)	0.09	126.21
Collings et al.	0.23	27.13
Pasandideh-Fard et al.	0.24	2.29
Jones (1971)	0.34	0.71
Choudhury et al. (analytic)	0.46	44.76
Chandra & Avedisian	0.61	1.19
Mao et al.	0.62	0.18
ANN (this work)	**0.78**	**0.02**

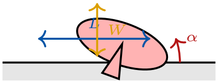
Sensitivity analysis (OVAT):

- r_a has the largest effect on model accuracy
- We has greater effect than Re (relevant at micro-scale)
- θ has the least influence

Key finding:
Omitting r_a (as done by most prior models) causes a **significant decrease** in R^2 and **increase** in RMSE — roughness is essential.

Traditional Ellipse Fitting (Balthazard, 1939)

A spherical drop projected onto a flat surface produces an ellipse. The **inverse-sine rule**:



$$\alpha = \arcsin\left(\frac{W}{L}\right)$$

Limitations (Joris et al., 2014):

- Ignores oscillations, air resistance, surface friction
- Assumes perfectly elliptical stain outline
- Accuracy degrades for $\alpha \in [20°, 50°]$
- Tail must be removed manually

Reported accuracy: 5–$7°$ (Bevel & Gardner); $\approx 3°$ for $\alpha \leq 45°$.

Active Bloodstain Shape Model (ABSM)

Data-driven pipeline (Joris et al., 2014):

Step 1 – Data collection: 367 stains at angles $5°$–$90°$ (step $2.5°$); pig blood; pipette at 1.2 m; 12 stains per angle.

Step 2 – Preprocessing: Karhunen–Loève (KL) transform $\rightarrow$ Otsu threshold $\rightarrow$ flood-fill $\rightarrow$ contour extraction $\rightarrow$ *manual* tail removal.

Step 3 – Landmark placement: Each contour resampled to 78 homologous landmark points via TPS-RPM non-rigid registration.

Step 4 – Shape superimposition: Generalised Procrustes analysis (GPA) removes location, scale, and rotation — leaves only shape.

Step 5 – PCA: Extract principal components from landmark cloud.

Step 6 – Regression: Link PC scores to impact angle.

Procrustes Superimposition

GPA minimises the least-squares difference between corresponding landmarks across all training stains, yielding a normalised **mean shape** and shape deviations.

PCA Result

First PC alone accounts for **97.7%** (later 98.4%) of all shape variation — stain shape is essentially one-dimensional.

TPS-RPM Registration Algorithm

Non-rigid image registration cast as a binary linear assignment + least-squares minimisation (Chui & Rangarajan, 2003):

$$\min_{Z,f} E(Z,f) = \min_{Z,f} \left[\sum_{i=1}^{N} \sum_{a=1}^{K} z_{ai} \left\| x_i - f(v_a) \right\|^2 + \lambda \| Lf \|^2 - \zeta \sum_{i=1}^{N} \sum_{a=1}^{K} z_{ai} \right]$$

- **Term 1:** minimise distance between target points x_i and transformed model points $f(v_a)$
- **Term 2:** smoothness regularisation (λ)
- **Term 3:** prevent over-rejection of points as outliers (ζ)

Fuzzy (soft-assign) version replaces binary Z with continuous $M \in [0,1]$ and adds entropy control via temperature T:

$$\min_{M,f} E(M,f) = \min_{M,f} \left[\sum_{i,a} m_{ai} \| x_i - f(v_a) \|^2 + \lambda \| Lf \|^2 - \zeta \sum_{i,a} m_{ai} + T \sum_{i,a} m_{ai} \ln m_{ai} \right]$$

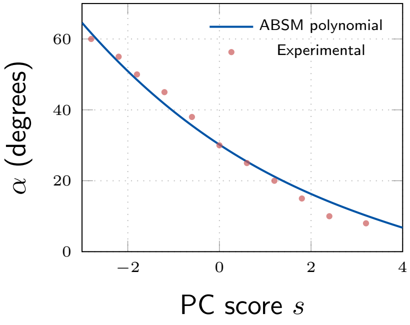
Polynomial Regression Model

Instead of $\arcsin(W/L)$, a **third-order polynomial** is fitted to link the first PC score s to impact angle α:

$$\alpha(s) = -0.05\,s^3 + 0.85\,s^2 - 8.46\,s + 30.2$$

Why polynomial instead of inverse-sine?

Width-to-length ratios *consistently underestimate* the inverse-sine for $\alpha \in [20°, 50°]$ — polynomial regression corrects this bias automatically.



ABSM Accuracy vs. Ellipse Fitting

Table: Accuracy comparison (Joris et al., 2014)

Method	RMSE	$\leq 2^\circ$ error
Auto ellipse fitting	2.09°	69%
ABSM	$\mathbf{1.44^\circ}$	**89%**

Mean regression residual (on training set): $\mathbf{1.03^\circ}$
Mean inverse-sine residual: 1.62°

Key advantages of ABSM:

- Fully automated — no inter/intra-observer variation
- Automatically ignores tail information
- Learns material-specific shape properties
- Extensible to multiple PC modes for different substrates

Degradation region

Both ABSM and ellipse model show highest errors for $\alpha \in [20^\circ, 50^\circ]$ — but ABSM recovers better due to polynomial (not inverse-sine) regression.

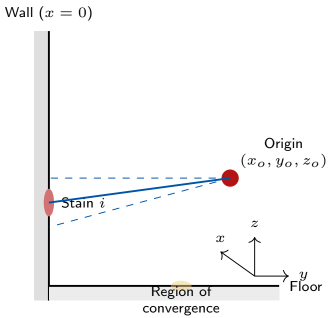
From 2-D Convergence to 3-D Origin

Traditional **method of strings**: assumes straight trajectories, gives only the *region of convergence* (2-D horizontal projection).

Known systematic error (Attinger et al., 2019): Straight-trajectory methods overestimate height by $\sim 50\%$ on average (Behrooz et al., 2011) — significant enough to confuse standing vs. sitting positions.

Proposed approach (Attinger et al., 2019):

1. Define a *range* of physically possible trajectories per stain (fluid dynamics)
2. Propagate measurement uncertainties statistically
3. Use maximum-likelihood to obtain a 3-D *confidence volume* for the origin



Equation of Motion for Blood Drops

Newton's law for a flying blood drop (Attinger et al., 2019):

$$m_d \frac{d\tilde{V}}{dt} = m_d \, \tilde{g} - \tilde{F}_D$$

Drag force:

$$\tilde{F}_D = \rho_a \, C_D \, \frac{A_d}{2} \, V \, \tilde{V}$$

where $A_d = \pi D^2/4$ (cross-section of undeformed drop), C_D corrected for drop deformation at intermediate Weber numbers.

Spread-factor correlation (Scheller & Bousfield, 1995; calibrated for cardstock):

$$\beta = \frac{W}{D} = a \, Re_N^{2b} \, Oh^{-b} \qquad (a = 0.257, \; b = 0.235)$$

Rearranged to give impact velocity u as a function of drop diameter D for a given β:

$$u = \left(\frac{\beta}{a}\right)^{1/(2b)} \frac{\mu^{1/2} \, \sigma^{1/4}}{(\rho D)^{3/4}}$$

Flight Breakup and Stain Shape Criteria

Criterion 1: Flight Breakup

A flying drop breaks when aerodynamic drag overcomes surface tension. Upper-bound constraint on trial velocity:

$$We_a = \frac{\rho_a \, D \, V^2}{\sigma} < K = 13$$

(Hsiang & Faeth, 1992)

Criterion 2: Stain Morphology

Stain periphery transitions (Bird et al., 2009):

$$\underbrace{We_n Re_n^{1/2}}_{\text{inertia vs. resistance}} \quad \text{vs.} \quad \underbrace{\frac{V_t}{V_n \sqrt{Re_n}}}$$

Spread-factor range: $\beta_{\text{trial}} \in [1.25,\ 6.0]$

For each trial β_{trial}:

1. Compute D_{trial} from $\beta = W/D$
2. Compute u_{trial} from spread correlation
3. Integrate Eq. of motion *backward*
4. Check criteria (1) and (2)
5. Keep only physically admissible trajectories

Result

Each stain produces a **bundle** of trial trajectories, not a single line.

Probabilistic Region of Origin

Step 1 — Region of convergence (horizontal plane). PDF of trajectory i passing point k:

$$\psi_{ik} = \frac{1}{\Delta\theta_i \sqrt{2\pi}} \exp\left(-\frac{\theta_{ik}^2}{2\Delta\theta_i^2} \right)$$

where $\theta_i = \dfrac{\pi}{2} + \arctan\left(\dfrac{\sin\gamma_i}{\tan\alpha_i} \right)$ and $\Delta\theta_i$ propagated from stain measurement errors.

Step 2 — Height estimation. Uniform PDF with Gaussian tails for height z above point k:

$$\phi_{ik}(z) = \begin{cases} A, & z_{\min} \le z \le z_{\max} \\ \dfrac{A}{\Delta z} \exp\left[-\frac{1}{2} \left(\dfrac{z_{\min} - z}{\Delta z} \right)^2 \right], & z < z_{\min} \\ \dfrac{A}{\Delta z} \exp\left[-\frac{1}{2} \left(\dfrac{z - z_{\max}}{\Delta z} \right)^2 \right], & z > z_{\max} \end{cases}$$

Step 3 — Joint likelihood. For N stains:

$$\psi_k = \prod_{i=1}^{N} \psi_{ik}, \qquad \phi_k(z) = \prod_{i=1}^{N} \phi_{ik}(z)$$

$$f(k, z) = B\, \psi_k \cdot \phi_k(z), \qquad P = \int_V f(k, z)\, dV$$

Reconstruction Results and Uncertainty Scaling

Experimental spatter patterns tested:

Pattern	x_0 (cm)	v_{imp} (m/s)	Type
HP 31	30	7.8	Fast/close
HP 7	60	5.2	—
HP 53	60	7.8	—
HP 11	120	5.2	—
HP 24	120	7.8	—
HP 21	190	7.8	—
C9	120	2.4	Slow/far

Key scaling law:

$$V_{\mathrm{RO}} \sim x_0^n, \qquad 5.2 \leq n \leq 5.5$$

Uncertainty grows as *fifth power* of source-to-wall distance because $\Delta x, \Delta y \sim x_0$ and $\Delta z \sim g x_0^2/(2u^2)$.

Selected results:

- HP 31 ($x_0 = 30$ cm): $V_{\mathrm{RO}}(P = 99.9\%) = \mathbf{0.11\ L}$ $\approx \frac{1}{10}$ of a grapefruit
- C9 ($x_0 = 120$ cm): $V_{\mathrm{RO}}(P = 99.9\%) = \mathbf{95\ L}$ ≈ 13 basketballs

Error analysis:

$|dx|, |dz| \leq 10$ cm for all 8 patterns.
No systematic bias — unlike the method of strings which *always* overestimates height.

Automated BPA Feature Extraction (Arthur et al., 2017)

Motivation: Current BPA taxonomy is *mechanism-focused*, leading to subjective analyst conclusions and inter-observer conflicts.

Goal: Automated, quantitative, *observation-first* pattern description.

Image-processing pipeline:

1. Pattern creation and digitisation
2. Background subtraction (median filter + KL contrast)
3. Element segmentation (Triangle threshold)
4. Cropping and binary image
5. Morphological erosion/dilation (tail removal)
6. Element labelling (**8-connected components**)
7. Feature extraction (local & global)

Test pattern facts:

- Porcine blood; impact device; 2.5 mL pool
- Nikon D810, 36.3 MP; 10.1 pix/mm
- 420 total elements detected
- 262 elements (62%) classified as elliptical
- Mean impact angle: $49.6^\circ \pm 17.3^\circ$
- Impact angle range: 12.2°–90°

Local Features: Shape and Orientation

Impact angle (Eq. 1)

$$\alpha = \arcsin\left(\frac{\text{Minor axis}}{\text{Major axis}}\right)$$

Tail-to-body ratio (Eq. 2)

$$\frac{\text{Length of tail}}{\text{Length of body} + \text{Length of tail}}$$

Mean: 0.21 (21% of element is tail).
Range: 0.02–0.85.
113 of 262 elliptical elements had measurable tails.

Convex hull irregularity (Eq. 3)

$$\text{Irreg}_1 = \frac{\text{Area of element}}{\text{Area of convex hull}}$$

Mean: 0.40 over all elements.

Inscribed-circle irregularity (Eq. 4)

$$\text{Irreg}_2 = \frac{\text{Perimeter of inscribed circle}}{\text{Perimeter of element body}}$$

Mean: 0.71; Range: 0.23–0.92.

Global Features: Pattern-Level Metrics

Linearity (polynomial fit, Eq. 5)

A 3rd-degree polynomial $f(x) = ax^3 + bx^2 + cx + d$ is fitted to the centroids of elliptical elements.

$R^2 = 0.26$ for the test impact pattern (low, as expected — elements radiate outward rather than linearly).

Mean distance to polynomial: 55.2 mm.

Gamma angle distribution

Direction	% of 420 elements
Upward-right	28.1%
Upward-left	9.8%
Downward-right	4.8%
Downward-left	5.2%
Horiz. right	7.1%
Horiz. left	7.4%

Std dev of orientation angle: 51.1°

(high = wide radiating spread, as expected).

Circularity of convex hull (Eq. 6)

$$C = \frac{4\pi A}{P^2}$$

Test pattern: $C = 0.79$ (nearly circular, as expected for an impact spatter).

Element density

Overall: 0.34 el/cm^2; max local: 1.3 el/cm^2

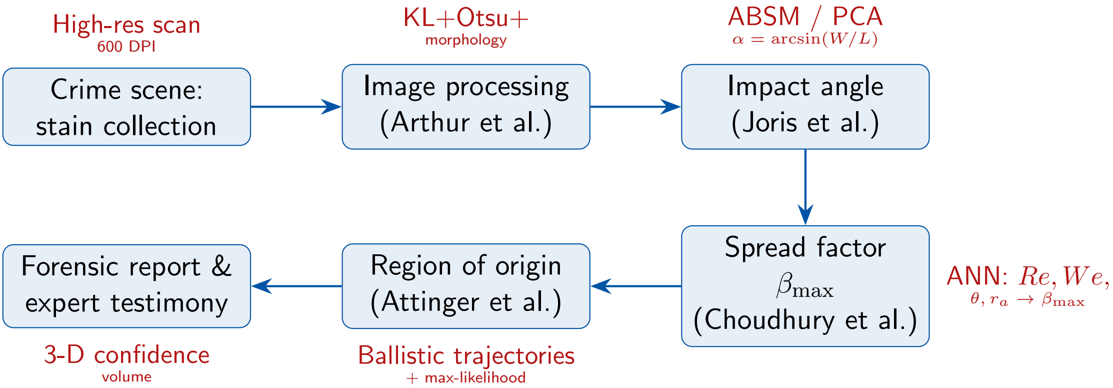
Connecting the Four Studies

Paper	Core problem	Key method	Key result
Joris et al. (2014)	Impact angle from stain shape	ABSM + PCA + polynomial regression	RMSE $1.44°$, 89% within $2°$
Choudhury et al. (2023)	Predict $\beta_{\max}$ incl. surface roughness	4-4-1 ANN (LM training)	$R^2 = 0.99$ (train), 0.78 (blind); MSE $=$ 0.02
Attinger et al. (2019)	3-D region of origin with uncertainty	Fluid dynamics + max-likelihood stats	Error ≤ 10 cm; $V_{\mathrm{RO}} \sim x_0^{5.3}$
Arthur et al. (2017)	Objective quantitative pattern features	Image processing (12 local/global features)	Measurable criteria for taxonomy

Common thread

All four papers replace *human subjectivity* with *physics-informed, data-driven objectivity* — a transition critical for BPA admissibility in forensic testimony.

The Unified BPA Workflow



Critical Observations and Open Problems

Limitations identified across papers:

- ABSM trained on pig blood on white paper — needs calibration per substrate and fluid
- ANN $\beta_{\max}$ model valid only within parameter bounds of training data ($We \leq 2055$, $Re \leq 15{,}860$)
- Attinger et al. model assumes quiescent air and non-interacting drops (not valid near gunshot origin)
- Image processing validated on a single pattern only
- Straight-trajectory assumption (method of strings) still used in practice despite 50% height error

Open research directions:

- Transfer learning to extend ABSM across fluids and surfaces without full re-training
- Physics-informed neural networks (PINNs) to replace purely empirical ANN for $\beta_{\max}$
- 3-D scanning + ABSM + Attinger pipeline into a single automated crime-scene software
- Uncertainty quantification for Arthur et al. features using bootstrap methods
- Human blood datasets (currently pig blood used as proxy)
- Extension of ABSM to $5°$–$90°$ range (currently $5°$–$60°$)

Surface Roughness: The Often-Ignored Variable

Sensitivity analysis (Choudhury et al., 2023) reveals:

$$r_a \succ We \succ Re \succ \theta$$

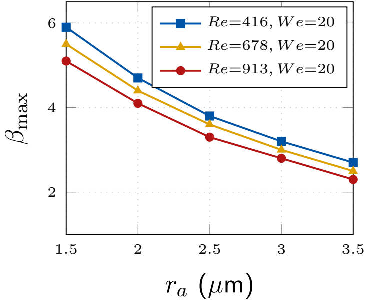
in terms of influence on $\beta_{\max}$.

Physical interpretation:

- Rough surfaces create **additional resistance** to lateral expansion
- Wenzel (1936): real surface contact angle modified as

$$\cos\theta^* = r_w \cos\theta_{\text{Young}}$$

  where r_w is the roughness ratio
- $\beta_{\max}$ **decreases** monotonically with increasing r_a (ANN correctly captures this)
- Impact on BPA: different flooring/wall surfaces



Illustrative: $\beta_{\max}$ vs. r_a (fixed We)

Summary of Key Equations

Quantity	Formula	Source/Comment
Impact angle (classical)	$\alpha = \arcsin(W/L)$	Balthazard (1939)
Impact angle (ABSM)	$\alpha(s) = -0.05s^3 + 0.85s^2 - 8.46s + 30.2$	Joris et al. (2014)
Spread factor	$\beta_{\max} = d_{\max}/d$	Definition
Weber number	$We = \rho V^2 d/\sigma$	—
Reynolds number	$Re = \rho V d/\mu$	—
Drop equation of motion	$m_d\dot{\tilde{V}} = m_d\tilde{g} - \tilde{F}_D$	Attinger et al. (2019)
Spread-factor correlation	$\beta = a\, Re_N^{2b} Oh^{-b}$	Scheller & Bousfield (1995)
Flight-breakup limit	$We_a = \rho_a D V^2/\sigma < 13$	Hsiang & Faeth (1992)
Uncertainty volume scaling	$V_{\mathrm{RO}} \sim x_0^{5.3}$	Attinger et al. (2019)
Tail/body ratio	$l_{\mathsf{tail}}/(l_{\mathsf{body}} + l_{\mathsf{tail}})$	Arthur et al. (2017)
Convex-hull circularity	$C = 4\pi A/P^2$	Arthur et al. (2017)

Conclusions

1. **Choudhury et al. (2023):** A 4-4-1 ANN trained on 342 data points outperforms all analytic and empirical models for $\beta_{\max}$ prediction ($R^2 = 0.78$ blind test vs. next best 0.62), primarily because it incorporates surface roughness r_a as an explicit input.
2. **Joris et al. (2014):** The ABSM replaces the inverse-sine rule with a data-driven polynomial regression on PCA scores, cutting RMSE from 2.09° to 1.44° and increasing $\leq 2^\circ$ accuracy from 69% to 89%.
3. **Attinger et al. (2019):** Ballistic reconstruction with probabilistic uncertainty propagation yields a 3-D confidence volume with error ≤ 10 cm and no systematic height bias, while uncertainty scales as $x_0^{\sim 5.3}$.
4. **Arthur et al. (2017):** A fully automated 12-feature image-processing pipeline provides objective, measurable pattern criteria absent from current BPA taxonomies.
5. **Overall:** Physics + machine learning + probabilistic reasoning together form a rigorous, reproducible BPA pipeline ready for forensic deployment.

References

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Thank You

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